Juliette Woodrow

Ph.D. Candidate in Computer Science, Stanford University

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Research Interests

Machine learning for education, interpretable representations of learners, probabilistic modeling and uncertainty in assessment, evaluating language models with people in the loop, and AI systems that create more opportunities for people to learn from one another. I build and evaluate these systems in deployed courses with tens of thousands of students and thousands of volunteer teachers.

Education

Stanford University — Stanford, CA
Ph.D. in Computer Science. Advisor: Chris Piech — *expected 2027*

Stanford University — Stanford, CA
M.S. in Computer Science — *2022*

Stanford University — Stanford, CA
B.S. in Computer Science — *2021*

Awards and Fellowships

- HAI Graduate Research Fellowship, Stanford Institute for Human Centered Artificial Intelligence — 2023
- Walter J. Gores Award for Excellence in Teaching, Stanford University — 2022
  Stanford's highest award for teaching, presented at University Commencement.
- Centennial Teaching Assistant Award, Stanford School of Engineering — 2022

Publications

* denotes equal contribution.

Conference Papers (peer reviewed)

[C1] **Juliette Woodrow** and Chris Piech. **PromptNCE: Conditional Probabilities and PMI Using Only LLMs and Contrastive Estimation Prompts.** *Conference on Language Modeling (COLM), 2026*

[C2] Charles Koutcheme, **Juliette Woodrow**, and Chris Piech. **Aligning Small Language Models for Programming Feedback: Towards Scalable Coding Support in a Massive Global Course.** *ACM Technical Symposium on Computer Science Education (SIGCSE), 2026*

[C3] **Juliette Woodrow**, Sanmi Koyejo, and Chris Piech. **Improving Generative AI Student Feedback: Direct Preference Optimization with Teachers in the Loop.** *International Conference on Educational Data Mining (EDM), 2025*

[C4] **Juliette Woodrow** and Chris Piech. **Soft Grades: A Calibrated and Accurate Method for Course Grade Estimation that Expresses Uncertainty.** *International Learning Analytics and Knowledge Conference (LAK), 2025*

[C5] Allen Nie, Yash Chandak, Miroslav Suzara, Ali Malik, **Juliette Woodrow**, Matt Peng, Mehran Sahami, Emma Brunskill, and Chris Piech. **The GPT Surprise: Offering Large Language Model Chat in a Massive Coding Class Reduced Engagement but May Increase Adopters' Exam Performances.** *ACM Conference on Learning at Scale (L@S), 2025*

[C6] Ali Malik*, **Juliette Woodrow***, Chao Wang, and Chris Piech. **TeachNow: Enabling Teachers to Provide Spontaneous, Realtime 1:1 Help in Massive Online Courses.** *ACM Conference on Innovation and Technology in Computer Science Education (ITiCSE), 2024*

[C7] **Juliette Woodrow**, Ali Malik, and Chris Piech. **AI Teaches the Art of Elegant Coding: Timely, Fair, and Helpful Style Feedback in a Global Course.** *ACM Technical Symposium on Computer Science Education (SIGCSE), 2024*

[C8] Ali Malik, **Juliette Woodrow**, and Chris Piech. **Learners Teaching Novices: An Uplifting Alternative Assessment.** *ACM Technical Symposium on Computer Science Education (SIGCSE), 2024*

[C9] Evan Zheran Liu, David Yuan, Ahmed Ahmed, Elyse Cornwall, **Juliette Woodrow**, Kaylee Burns, Allen Nie, Emma Brunskill, Chris Piech, and Chelsea Finn. **A Fast and Accurate Machine Learning Autograder for the Breakout Assignment.** *ACM Technical Symposium on Computer Science Education (SIGCSE), 2024*

Under Review

[R1] **Juliette Woodrow**, Zikun Zhu, and Chris Piech. **Learner Insight Tracing: Natural Language Knowledge States that Predict Future Work.** *Under review, 2026*

Preprints and Technical Reports

[P1] Ali Malik*, **Juliette Woodrow***, Brahm Capoor, Thomas Jefferson, Miranda Li, Sierra Wang, Patricia Wei, Dora Demszky, Jennifer Langer-Osuna, Julie Zelenski, Mehran Sahami, and Chris Piech. **Code in Place 2023: Understanding Learning and Teaching at Scale through a Massive Global Classroom.** *Technical report, 2023*

Curricular Contributions (peer reviewed)

[N1] Chris Piech, Mehran Sahami, Yasmine Alonso, Katie Liu, Javokhir Arifov, Anjali Sreenivas, Dan Webber, Tina Zheng, Ngoc Nguyen, Iddah Mlauzi, and **Juliette Woodrow**. **Infinite Story.** *SIGCSE Nifty Assignments, 2025*

[N2] **Juliette Woodrow**, Katie Creel, and Nick Bowman. **Bias Bars.** *SIGCSE Nifty Assignments, 2022*

Research Experience

Graduate Researcher, Stanford AI Lab 2022 – present

Advisor: Chris Piech *Stanford, CA*

- Learner Insight Tracing. Built and deployed an LLM pipeline that turns students' code histories into natural language summaries of their understanding, evaluated by how well they predict future work. Deployed for 5,449 students and 451 volunteer teachers.
- PromptNCE. Introduced a training free method for estimating conditional probabilities and pointwise mutual information using only prompted language models.
- Teacher preference optimization. Fine tuned Llama 3.1 8B with DPO on teachers' grading choices and edits. Experts preferred its feedback over GPT-4o in a blind study.
- Soft Grades. Developed an item response model that represents ability and grades as calibrated probability distributions.
- TeachNow. Designed and built real time matching of volunteer teachers with students, supporting 15,000+ sessions and 4,000+ hours of 1:1 help.

Teaching

Co-Creator and Co-Instructor, Code in Place X 2026 – present

Stanford University *Global, online*

- Co-designing and teaching an experimental CS1 course for 3,000+ students in which AI use is allowed on every assignment and understanding is assessed through weekly live oral assessments with a teacher and peers.
- Built teacher training with simulated students, AI summaries of each student's prior work for their section teacher, and weekly surveys of how students use AI.

Instructor of Record, CS109: Probability for Computer Scientists Winter 2026

Stanford University *Stanford, CA*

- Sole instructor for a required core course of 270+ students. Delivered all lectures, managed 10 TAs, and wrote all exams.
- Taught probability through computation, connecting conditional probability, inference, and information theory to language models and diffusion models.
- Contributing author to the online course reader, adding sections on calling LLM APIs and an interactive computational inference demonstration.

Instructor of Record, CS106A: Programming Methodology Summer 2021

Stanford University *Stanford, CA*

- Taught Stanford's introductory Python course to 120+ students. Delivered all lectures, designed assessments, and managed course staff.

Founding Team, Code in Place 2020 – present

Stanford University *Global, online*

- Helped build and run a free global introductory programming course that has served 60,000+ students with 5,000+ volunteer teachers.
- Wrote weekly section materials, delivered lectures, trained teaching staff, and built platform infrastructure for course delivery, assessment, and teacher operations.

Teaching Assistant 2020 – 2025
Stanford University *Stanford, CA*

- Head TA, CS109 (Fall 2025). TA, CS109 (Fall 2023). Head TA, CS106A (Fall 2020 to Spring 2022).

Mentoring

- Josh Delgadillo, Learner Insight Tracing Fall 2026 – present
- Zikun Zhu, Learner Insight Tracing (coauthor)
- Trained and supported thousands of volunteer section leaders in Code in Place, and teaching assistants in CS106A and CS109.

Service

- Reviewer, ACM Technical Symposium on Computer Science Education (SIGCSE)
- Code in Place, volunteer teacher recruitment and training for a global teaching community 2020 – present

Proposed Service and Activities

- Open, free online courses at scale that pair AI with human teachers, building on Code in Place.
- Pathways for undergraduates to become teachers and researchers through peer teaching and course staff roles.

Technical Skills

Machine learning, probabilistic modeling, item response theory, LLM system design and evaluation, preference optimization, randomized and quasi experimental evaluation, Python, PyTorch, Hugging Face, TypeScript, React, and full stack web development.